Homework Assignment 1 — Solution

Notation: two risky assets 1, 2 and a risk-free asset with return r_f. $\mu_i^e = \mu_i - r_f$, $\sigma_{12} = \rho_{12}\sigma_1\sigma_2$. Excel file: hw1_frontier.xlsx (all cells are live formulas; inputs on the *Parameters* sheet).

> **Assumption.** The problem lists $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho_{12}$ but not r_f. I use $r_f = 0.05$. Change cell `Parameters!B9` if your class used a different value; every number and chart updates.

Problem 1 — Tangency portfolio

Step 1: Unconstrained optimization

With $r_p = w_1 r_1 + w_2 r_2$,

$$\mu_p^e = w_1\mu_1^e + w_2\mu_2^e, \qquad \sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2\sigma_{12}.$$

(If the weights do not sum to one, the leftover $1 - w_1 - w_2$ sits in the risk-free asset, so the excess return is still $\sum_i w_i \mu_i^e$.)

Maximize $S(w_1, w_2) = \mu_p^e/\sigma_p$. By the quotient rule,

$$\frac{\partial S}{\partial w_i} = \frac{\dfrac{\partial \mu_p^e}{\partial w_i}\sigma_p - \mu_p^e\dfrac{\partial \sigma_p}{\partial w_i}}{\sigma_p^2} = 0 \;\Rightarrow\; \frac{\partial \mu_p^e}{\partial w_i}\sigma_p = \mu_p^e\frac{\partial \sigma_p}{\partial w_i}.$$

Since $\sigma_p = \sqrt{\sigma_p^2}$, $\dfrac{\partial \sigma_p}{\partial w_i} = \dfrac{1}{2\sigma_p}\dfrac{\partial \sigma_p^2}{\partial w_i}$. Substituting and multiplying both sides by σ_p:

$$\frac{\partial \mu_p^e}{\partial w_i}\sigma_p^2 = \mu_p^e\,\frac{1}{2}\frac{\partial \sigma_p^2}{\partial w_i}, \qquad i = 1, 2.$$

Dividing the $i = 1$ condition by the $i = 2$ condition, the common factors σ_p^2 and $\frac{1}{2}\mu_p^e$ cancel:

$$\frac{\partial \mu_p^e/\partial w_1}{\partial \mu_p^e/\partial w_2} = \frac{\partial \sigma_p^2/\partial w_1}{\partial \sigma_p^2/\partial w_2} \quad\Leftrightarrow\quad \frac{\partial \mu_p^e/\partial w_1}{\partial \sigma_p^2/\partial w_1} = \frac{\partial \mu_p^e/\partial w_2}{\partial \sigma_p^2/\partial w_2}.$$

Economic interpretation. Treat expected excess return as the "good" the investor wants and variance as the "cost" (or "bad") paid for it. Each asset is an input that produces both.

- The left side is the rate at which shifting money between asset 1 and asset 2 changes expected return; the right side is the rate at which the same shift changes variance. At the optimum the two rates are equal, the same way a consumer's MRS equals the price ratio, or a firm's MRTS equals the factor-price ratio.
- The second form is the **equimarginal principle** ($MU_1/p_1 = MU_2/p_2$): the marginal expected excess return per unit of marginal variance must be the same for both assets. If asset 1 gave more extra return

per unit of extra risk than asset 2, moving a little wealth from 2 to 1 would raise the Sharpe ratio, so the starting point could not be optimal.

- $\frac{\partial \sigma_p^2}{\partial w_i} = 2\,\mathrm{Cov}(r_i, r_p)$, so the condition reads $\mu_i^e / \mathrm{Cov}(r_i, r_p)$ equal across assets. Each asset's reward is proportional to its marginal contribution to portfolio risk, which is the beta pricing relation $\mu_i^e = \beta_{ip}\,\mu_p^e$.
- S is homogeneous of degree 0 in (w_1, w_2) (doubling both weights leaves the Sharpe ratio unchanged). The first-order conditions therefore pin down only the **ratio** w_1/w_2, i.e. the direction of the ray, not its scale. That is why Step 2 is needed.

Step 2: Impose the budget constraint

The partial derivatives are

$$\frac{\partial \mu_p^e}{\partial w_1} = \mu_1^e, \quad \frac{\partial \mu_p^e}{\partial w_2} = \mu_2^e, \quad \frac{\partial \sigma_p^2}{\partial w_1} = 2(w_1\sigma_1^2 + w_2\sigma_{12}), \quad \frac{\partial \sigma_p^2}{\partial w_2} = 2(w_2\sigma_2^2 + w_1\sigma_{12}).$$

The ratio condition gives

$$\frac{\mu_1^e}{\mu_2^e} = \frac{w_1\sigma_1^2 + w_2\sigma_{12}}{w_2\sigma_2^2 + w_1\sigma_{12}} \;\Rightarrow\; \mu_1^e(w_2\sigma_2^2 + w_1\sigma_{12}) = \mu_2^e(w_1\sigma_1^2 + w_2\sigma_{12})$$

$$\Rightarrow\; w_1(\mu_2^e\sigma_1^2 - \mu_1^e\sigma_{12}) = w_2(\mu_1^e\sigma_2^2 - \mu_2^e\sigma_{12}).$$

Now impose $w_1 = w, w_2 = 1 - w$:

$$w(\mu_2^e\sigma_1^2 - \mu_1^e\sigma_{12}) = (1-w)(\mu_1^e\sigma_2^2 - \mu_2^e\sigma_{12})$$

$$w[\mu_1^e\sigma_2^2 + \mu_2^e\sigma_1^2 - (\mu_1^e + \mu_2^e)\sigma_{12}] = \mu_1^e\sigma_2^2 - \mu_2^e\sigma_{12}$$

$$w = \frac{\mu_1^e\sigma_2^2 - \mu_2^e\sigma_{12}}{\mu_1^e\sigma_2^2 + \mu_2^e\sigma_1^2 - (\mu_1^e + \mu_2^e)\sigma_{12}} \qquad \text{(Equation 1)} \;\blacksquare$$

Problem 2 — Numerical part ($\mu_1 = 0.10,\ \mu_2 = 0.14,\ \sigma_1 = 0.15,\ \sigma_2 = 0.20,\ \rho_{12} = 0.2,\ r_f = 0.05$)

Quantity	Value
$\sigma_{12} = 0.2(0.15)(0.20)$	0.006
$\mu_1^e,\ \mu_2^e$	0.05, 0.09
Tangency weights $w_1 = w,\ w_2 = 1 - w$	**0.4584, 0.5416**
μ_p	0.12166
$\sigma_p^2,\ \sigma_p$	0.019440, 0.13943
Sharpe ratio $(\mu_p - r_f)/\sigma_p$	0.5140
GMV portfolio: $w_1,\ \mu,\ \sigma^2,\ \sigma$	0.6733, 0.11307, 0.017109, 0.13080

Calculation of w: $w = \dfrac{0.05(0.04) - 0.09(0.006)}{0.05(0.04) + 0.09(0.0225) - 0.14(0.006)} = \dfrac{0.00146}{0.003185} = 0.4584.$

(a) Mean–standard deviation frontier (sheet *Charts*, chart 1)

- **Risky assets only:** for $w_1 \in [-0.5, 1.5]$, $\mu_c = w_1\mu_1 + (1-w_1)\mu_2$, $\sigma_c = \sqrt{w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\sigma_{12}}$. This traces a hyperbola.
- **With the risk-free asset:** the two straight lines $\mu_c = r_f \pm SR \cdot \sigma_c$ with $SR = 0.5140$. The upper line is the capital market line, tangent to the hyperbola at $p = (0.1394,\ 0.1217)$, where the portfolio holds 45.84% in asset 1 and 54.16% in asset 2.

(b) Mean–variance frontier (chart 2)

Risky assets only, analytic parabola. From $\mu_c = w\mu_1 + (1-w)\mu_2$, $w = \dfrac{\mu_2 - \mu_c}{\mu_2 - \mu_1}$ and $1 - w = \dfrac{\mu_c - \mu_1}{\mu_2 - \mu_1}$. Substituting into σ_c^2:

$$\sigma_c^2 = \frac{(\mu_2 - \mu_c)^2\sigma_1^2 + (\mu_c - \mu_1)^2\sigma_2^2 + 2(\mu_2 - \mu_c)(\mu_c - \mu_1)\sigma_{12}}{(\mu_2 - \mu_1)^2} = \sigma_{GMV}^2 + a(\mu_c - \mu_{GMV})^2,$$

with $a = \dfrac{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}{(\mu_2 - \mu_1)^2} = \dfrac{0.0505}{0.0016} = 31.5625$, $\mu_{GMV} = 0.11307$, $\sigma_{GMV}^2 = 0.017109$. So σ_c^2 is a quadratic in μ_c: a parabola opening to the right in (σ^2, μ) space. On the *Frontier* sheet, column E (σ^2 from the weights) and column F (the analytic formula) agree to machine precision (column G ≈ 0). In chart 2 the dots (weight grid) sit exactly on the analytic curve.

With the risk-free asset — derivation of Eq. (2). From $\mu_c = r_f + w_p(\mu_p - r_f)$ we get $w_p = \dfrac{\mu_c - r_f}{\mu_p - r_f}$. From $\sigma_c^2 = w_p^2\sigma_p^2$ we get $w_p = \pm\dfrac{\sqrt{\sigma_c^2}}{\sigma_p}$ (positive for a long position in p, negative for a short position). Equating the two:

$$\frac{\mu_c - r_f}{\mu_p - r_f} = \pm\frac{\sqrt{\sigma_c^2}}{\sigma_p} \Rightarrow \mu_c = r_f \pm \frac{\mu_p - r_f}{\sigma_p}\sqrt{\sigma_c^2}. \qquad \text{(Equation 2) } \blacksquare$$

So the frontier is a square-root function of σ_c^2, not a straight line. Chart 2 shows that the orange curve $0.05 + 0.5140\sqrt{\sigma^2}$ passes through $p = (0.01944,\ 0.1217)$ and touches the risky parabola there, staying above it everywhere else.

(c) Challenging question: ray vs tangent in (σ^2, μ) space (chart 3, sheet *Slopes*)

(a) Slope of the ray from $(0, r_f)$ to (σ_p^2, μ_p):

$$\text{slope}_{\text{ray}} = \frac{\mu_p - r_f}{\sigma_p^2} = \frac{0.07166}{0.019440} = 3.686.$$

(b) Tangent slope of the risky frontier at p. The Eq. (2) curve is tangent to the risky frontier at p, so the two share a slope there. Differentiating Eq. (2), upper branch:

$$\frac{d\mu_c}{d\sigma_c^2} = \frac{\mu_p - r_f}{\sigma_p} \cdot \frac{1}{2\sqrt{\sigma_c^2}} \xrightarrow{\sigma_c^2 = \sigma_p^2} \frac{\mu_p - r_f}{2\sigma_p^2} = 1.843.$$

As a check, differentiating the parabola directly gives $\frac{d\mu}{d\sigma^2} = \frac{\mu_1 - \mu_2}{2[w\sigma_1^2 - (1-w)\sigma_2^2 + (1-2w)\sigma_{12}]} = 1.843$ at $w = 0.4584$.

(c) Relationship:

$$\text{slope}_{\text{tangent at } p} = \tfrac{1}{2}\,\text{slope}_{\text{ray}}.$$

Unlike mean–SD space, where the CML ray *is* the tangent, in mean–variance space the tangent at p is exactly half as steep as the ray from r_f.

(d) Using it to find p. The tangency portfolio is the unique point on the efficient part of the risky frontier where the frontier's slope $d\mu/d\sigma^2$ equals one-half of the slope of the ray from $(0, r_f)$:

$$\frac{d\mu}{d\sigma^2}\Big|_p = \frac{1}{2} \cdot \frac{\mu_p - r_f}{\sigma_p^2}.$$

Geometrically, the tangent line at p hits the vertical axis at $\mu_p - \tfrac{1}{2}(\mu_p - r_f) = \frac{r_f + \mu_p}{2}$, the **midpoint** between r_f and μ_p (here 0.0858). To locate p, move along the frontier until its tangent line crosses the μ-axis halfway between r_f and the point's own mean.

Note that p is **not** the point that maximizes the ray slope $(\mu - r_f)/\sigma^2$. In the table below, the ray slope is still rising past p. Maximizing $(\mu - r_f)/\sigma^2$ is a different problem from maximizing the Sharpe ratio.

Excel check (sheet *Slopes*):

w_1	μ	σ^2	**Ray slope**	**Tangent slope**	**Tangent ÷ Ray**
0.00	0.1400	0.04000	2.250	0.588	0.261
0.20	0.1320	0.02842	2.885	0.837	0.290
0.40	0.1240	0.02088	3.544	1.449	0.409
0.4584 (p)	**0.1217**	**0.01944**	**3.686**	**1.843**	**0.500**
0.50	0.1200	0.01863	3.758	2.286	0.608
0.60	0.1160	0.01738	3.797	5.405	1.423
1.00 (inefficient)	0.1000	0.02250	2.222	−1.212	−0.545

Among the points checked, the ratio equals exactly ½ only at the tangency portfolio. Below p it is less than ½. Above p it is greater than ½ and blows up near the GMV point, where the frontier becomes vertical. On the inefficient branch it is negative.